A Major Project Report
(Phase-III)

On

Short-Term Solar Energy Generation Forecasting Using Weather Data and Machine Learning Model

Submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IILM UNIVERSITY

Session 2026-27

in

Computer Science and Engineering

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May, 2026

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CANDIDATE'S DECLARATION

We hereby certify that the work presented in the **Phase-II report** of the project entitled "**Short-Term Solar Energy Generation Forecasting Using Weather Data and Machine Learning Model**", in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in the **School of Computing Science and Engineering, IILM University**, Greater Noida, is an original work carried out by us under the supervision of **Ms. Kirti**, School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-III report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

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Introduction and Objectives

The global transition towards renewable energy necessitates accurate short-term solar power generation forecasting to mitigate grid instability caused by weather-dependent intermittency. This Phase III implementation report marks the critical transition from the theoretical frameworks and system architecture designed in Phase II into concrete, production-ready software engineering. Utilizing the DKASC Alice Springs dataset, this phase executes the rigorous coding, model training, and system validation required to bring the conceptual predictive modeling design to life, moving beyond foundational research to build a mathematically sound tool for mapping non-linear environmental variables to active power output.

The primary objective of Phase III is to develop a robust, modular predictive pipeline that adheres strictly to professional software development standards, migrating away from fragmented experimental scripts into a cohesive, object-oriented architecture. Furthermore, this phase aims to execute a rigorous performance analysis comparing two advanced tree-based machine learning algorithms—Random Forest and XGBoost—using strict chronological data partitioning to prevent look-ahead bias. Ultimately, this report validates the successful implementation of a highly accurate forecasting system, establishing the definitive groundwork for the final prototype demonstration and expert review.

System Design, Architecture, and Modular Integration

To adhere to professional coding standards and ensure the scalability of the predictive system, the codebase was entirely refactored. The initial linear experimental scripts developed in Google Colab were decoupled into a highly modular, object-oriented software architecture. This separation of concerns ensures that data processing, model training, and execution logic are distinct, easily maintainable, and verifiable.

The system integration relies on three core Python modules:

- **data_preprocessing.py (Data Pipeline and Feature Engineering):** Time-series forecasting requires meticulous data handling. This module is responsible for dataset ingestion, missing value treatment, and ensuring strict chronological sorting. Furthermore, it handles complex feature engineering. Solar irradiance is inherently cyclical, driven by the rotation of the Earth and its orbit around the sun. To allow the machine learning algorithms to mathematically interpret this periodicity, the module engineer's cyclic temporal features using sine and cosine transformations for both the hour of the day and the month of the year. This prevents the model from interpreting 'Hour 23' and 'Hour 0' as being far apart mathematically, preserving the continuous nature of time.
- **model_training.py (Algorithmic Encapsulation):** This module encapsulates the core machine learning logic. It handles the instantiation, hyperparameter configuration, and training execution for the two chosen tree-based regression algorithms: Random Forest and XGBoost. To facilitate deployment and prevent the need for continuous retraining, this module utilizes the joblib library to serialize and save the fully trained models as .pkl files, making the system ready for real-world integration.
- **main.py (System Orchestration):** Acting as the central nervous system of the project, this execution script orchestrates the entire pipeline. It calls the preprocessing functions, feeds the structured data into the training module, generates predictions on the validation set, calculates the final error metrics, and renders the milestone visualizations.

Testing, Validation Methodology, and Data Partitioning

The predictive models were trained and rigorously tested using the DKASC Alice Springs solar dataset. In standard machine learning classification tasks, random shuffling of data during the train-test split is standard practice. However, for time-series forecasting, random shuffling introduces "look-ahead bias" or data leakage, where the model learns from future events to predict past occurrences.

To maintain the temporal integrity and physical reality of the dataset, a strict, unshuffled chronological 80/20 train-validation split was enforced:

- **Training Set (80%):** 153,900 observations were utilized to train the models, allowing them to learn the historical, non-linear relationships between meteorological variables and power output.
- **Validation Set (20%):** 38,475 completely unseen, future observations were reserved strictly for testing. This ensures that the performance metrics reflect the model's true capability to generalize to new, real-world data.

Rigorous Performance Analysis and Algorithmic Comparison

The validation phase involved a head-to-head performance analysis of the two advanced tree-based regression models. The evaluation utilized three primary metrics: Root Mean Squared Error (RMSE) to penalize large forecasting errors, Mean Absolute Error (MAE) to measure the average magnitude of errors, and the Coefficient of Determination (R^2) to determine the proportion of variance explained by the model.

Quantitative Results Summary

Model Metric	Random Forest	XGBoost	Performance Delta
Coefficient of Determination (R^2)	0.9651	0.9627	+0.0024 (RF Superior)
Root Mean Squared Error (RMSE)	1.5450 kW	1.5968 kW	-0.0518 kW (RF Superior)
Mean Absolute Error (MAE)	0.7838 kW	0.8090 kW	-0.0252 kW (RF Superior)

Theoretical Interpretation and Conclusion

Both algorithms demonstrated exceptional predictive capability, successfully explaining over 96% of the variance in short-term active power generation. However, the quantitative analysis reveals that the Random Forest model marginally but consistently outperformed the XGBoost model across all three metrics.

This performance differential can be attributed to the architectural differences between the algorithms. Random Forest utilizes a "Bagging" (Bootstrap Aggregating) methodology, constructing hundreds of independent decision trees in parallel and averaging their outputs. This parallelization makes the algorithm highly robust and resilient to the inherent noise, sudden cloud cover fluctuations, and high variance typically found in raw meteorological data.

Conversely, XGBoost utilizes a sequential "Boosting" technique, where each subsequent decision tree is built to correct the residual errors of the previous ones. While generally a highly powerful approach, boosting algorithms can sometimes become overly sensitive to outliers or extreme fluctuations in the training data. In the context of this specific solar irradiance dataset, the XGBoost model slightly over-corrected for atmospheric noise, resulting in marginally higher error margins (an RMSE of 1.5968 kW compared to Random Forest's 1.5450 kW). Therefore, the Random Forest model is identified as the superior and more stable algorithm for this specific forecasting deployment.

Visual Validation and Milestone Completion

To support the quantitative metrics, the pipeline generates visual validations that confirm the physical accuracy of the machine learning models.

Feature Importance Analysis

The XGBoost feature importance analysis confirms that the predictive accuracy is driven by sound meteorological and temporal variables rather than random noise. The feature importance plot verifies that the engineered cyclic temporal features (such as the sine of the hour) and key environmental sensors (like direct irradiance and panel temperature) hold the highest relative importance, proving the model successfully mapped the physical realities of solar generation.

Actual vs. Predicted Power Generation

A plotting of the Actual vs. Predicted generation over a condensed 40-hour temporal window provides visual proof of the model's fidelity. The resulting curves demonstrate that the algorithms successfully map the diurnal solar generation cycles. The predicted curves tightly hug the actual generation data during peak daytime production hours. Crucially, the model correctly identifies nighttime periods, regressing the predicted active power directly to zero without any predictive latency, lag, or phantom generation.

Conclusion

The Phase III implementation successfully transformed the initial conceptual design into a fully functional, highly optimized machine learning pipeline characterized by modularity and clean system integration. Through rigorous testing with the DKASC Alice Springs dataset, both the Random Forest and XGBoost algorithms demonstrated exceptional performance, each explaining over 96% of the variance in short-term active power generation. The strict enforcement of a chronological train-validation split ensured the preservation of temporal data integrity, guaranteeing that the predictive models were evaluated on their true ability to generalize to unseen meteorological conditions without data leakage.

The comprehensive performance analysis ultimately identified the Random Forest algorithm as the superior model for this specific forecasting deployment. Its underlying bagging architecture proved more resilient to atmospheric noise and sudden high variance in the solar dataset compared to XGBoost's sequential boosting methodology, resulting in consistently lower Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). With the software implementation, rigorous testing, and algorithmic validation completely achieved, the predictive pipeline stands as a mathematically robust tool, fully prepared for the upcoming Project Exhibition phase and technical evaluation.